

CLANTON 2 NE, AL, USA

WMO: 750020

Lat: 32.8516N

Lon: 86.6115W

Elev: 178

StdP: 99.20

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 63891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.8	(c) -3.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 7.5	(k) 14.1	(l) 6.8	(m) 12.7	(n) 2.0	(o) 0	(p) 0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 34.8	(d) N/A	(e) 33.5	(f) N/A	(g) 32.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.7	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.4	(b) 5.6	(c) 5.0	(e) -9.1	(f) 35.9	(g) 3.1	(h) 1.6	(i) -11.3	(j) 37.1	(k) -13.1	(l) 38.0	(m) -14.9	(n) 38.9	(o) -17.1	(p) 40.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.7	9.7	13.2	17.1	21.4	25.4	26.5	26.4	23.9	17.8	11.5	9.2		
	DBStd	8.03	5.75	5.46	4.89	3.66	3.10	1.85	1.66	2.00	2.81	4.33	4.43	5.21		
	HDD10.0	350	137	68	26	2	0	0	0	0	0	2	33	82		
	HDD18.3	1413	362	246	169	65	13	0	0	0	2	63	209	284		
	CDD10.0	3063	33	59	124	216	352	463	511	509	417	243	78	58		
	CDD18.3	1084	1	3	9	28	107	213	252	251	168	46	3	2		
	CDH23.3	9674	1	10	67	271	919	1979	2312	2291	1397	405	19	3		
	CDH26.7	3939	0	0	6	33	279	878	1018	1053	568	102	2	0		
(14) Wind	WSAvg	2.1	2.4	2.6	2.7	2.4	2.0	1.8	1.8	1.8	1.8	2.0	2.0	2.4		
(15) Precipitation	PrecAvg	1363	138	136	147	104	98	118	124	96	94	78	117	119		
	PrecMax	2022	256	245	288	183	232	259	324	199	226	229	241	236		
	PrecMin	779	71	48	31	30	12	34	36	23	9	13	37	67		
	PrecStd	241	51	50	75	39	59	65	62	46	60	56	53	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	25.0	27.7	29.0	32.1	35.2	35.6	37.6	35.2	31.2	25.7	23.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	20.2	22.7	25.3	27.7	30.7	33.7	33.8	35.1	33.3	29.1	23.2	21.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	18.2	20.4	23.3	26.5	29.6	32.6	32.8	33.5	31.7	27.5	21.5	20.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	16.1	18.7	21.5	25.0	28.3	31.5	31.7	31.8	30.2	25.5	19.9	18.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.2	12.7	11.8	11.1	10.3	10.6	11.3	12.8	13.2	10.4		
		MCDBR	12.1	12.4	13.4	14.0	13.9	13.0	11.7	13.0	13.6	14.3	14.1	11.4		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.325	0.356	0.387	0.443	0.472	0.526	0.521	0.427	0.367	0.341	0.321		
	taud		2.532	2.512	2.436	2.336	2.226	2.187	2.051	2.056	2.307	2.461	2.476	2.538		
	Ebn at Noon		904	932	922	902	851	822	776	773	838	871	866	880		
	Edh at Noon		84	94	109	126	142	148	168	165	123	98	88	79		
(44) All-Sky Solar Radiation	RadAvg		2.72	3.27	4.42	5.58	6.13	6.17	5.98	5.54	4.88	4.11	3.07	2.32		
	RadStd		0.20	0.41	0.41	0.39	0.45	0.46	0.41	0.31	0.42	0.52	0.31	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+116

Nomenclature: See separate page